

MA546 Introduction to Quantitative Finance

Assignment 3

1. What is **binary encoding**? Explain how portfolio weights or investment decisions can be represented using binary strings.
2. What is **Gray code**? Explain how it is constructed from binary representation and why it might be used in optimization of portfolios.
3. What is the key **difference between binary encoding and Gray encoding**? Discuss in terms of:
 - (a) bit changes between consecutive numbers,
 - (b) Hamming distance,
4. Convert the following **binary strings to Gray code** and compute their integer values, representing portfolio allocation levels:

Solution	Binary	Gray	Integer (Base 10)
S_1	000101		
S_2	001111		
S_3	010010		

5. Convert the following **Gray code to binary** and compute their integer values:

Solution	Gray	Binary	Integer (Base 10)
S_4	111010		
S_5	101101		
S_6	010111		

6. Portfolio Allocation, Fitness, and Roulette Selection

Solution	Allocation x (%)	Fitness $f(x) = x^2$	Probability	Cumulative Probability	Selected Solution (r)
S_1	10				0.12
S_2	25				0.35
S_3	30				0.58
S_4	45				0.77
S_5	50				0.89
S_6	60				0.95

- (a) Compute the fitness $f(x) = x^2$ for each portfolio allocation.
- (b) Compute the selection probability of each solution.
- (c) Calculate the cumulative probability for roulette-wheel selection.
- (d) Using the given r values in the last column, determine the selected solution for each random number.
- (e) Perform tournament selection (size = 2):

Tournament	Participants	Winner
T_1	(S_1, S_4)	
T_2	(S_2, S_5)	
T_3	(S_3, S_6)	
T_4	(S_1, S_3)	
T_5	(S_2, S_4)	
T_6	(S_5, S_6)	

7. Crossover and Mutation for binary GA

- (a) One-point crossover at position 3:

$$P_1 = 101101, \quad P_2 = 011010$$

Chromosome	Binary
O_1	
O_2	

- (b) Two-point crossover (positions 2 to 5):

$$P_1 = 110011, \quad P_2 = 001101$$

Chromosome	Binary
O_1	
O_2	

(c) Uniform crossover with mask $M = 110010$:

$$P_1 = 101010, \quad P_2 = 010101$$

Offspring	Binary
O	

(d) Identify parents given offspring:

$$O_1 = 101010, \quad O_2 = 011101$$

(Assume one-point crossover at position 3)

Parent	Binary
P_1	
P_2	

(e) Perform crossover and mutation:

$$P_1 = 111000, \quad P_2 = 000111$$

Crossover at position 3, mutation at bit 2.

Stage	Binary
After crossover	
After mutation	

8. Real-Coded GA and Inverse CDF Sampling

- (a) Explain the **inverse transform method** for generating random variables. How can this be used to generate offspring in a real-coded GA?
- (b) Suppose portfolio weights $x \in [0, 1]$ follow a probability distribution with cumulative distribution function (CDF):

$$F(x) = x^2$$

Generate 5 random offspring using uniform random numbers $u = [0.1, 0.25, 0.5, 0.75, 0.9]$ by applying the inverse CDF method. Fill the table:

Uniform Random u	Offspring $x = F^{-1}(u)$	Notes/Observation
0.1		
0.25		
0.5		
0.75		
0.9		

- (c) Consider the offspring generation using **polynomial mutation** in real-coded GA. The mutation is defined as:

$$x' = x + \delta(x_{max} - x_{min}), \quad \delta = (2u)^{1/(n_m+1)} - 1, \quad u \sim U(0, 1)$$

For $x = 0.4$, $x_{min} = 0$, $x_{max} = 1$, $n_m = 2$, generate 3 mutated offspring for $u = 0.2, 0.5, 0.8$. Fill the table:

Random u	δ	Mutated Offspring x'
0.2		
0.5		
0.8		